

Hypothesis Testing

Statistics Notes - Definitions, Formulas, Tests, Examples, and Exam Format

These notes are prepared from your uploaded Hypothesis Testing PDF in the same easy exam-friendly style we discussed earlier.

1. Statistical Inference

Statistical inference means using sample data to make conclusions about a larger population.

- Population: The entire group we want to study. Example: all customers of a company.
- Sample: A smaller selected part of the population. Example: 100 customers selected for a survey.
- Parameter: A numerical value describing the population, such as population mean μ , variance σ^2 , and standard deviation σ .
- Statistic: A numerical value describing the sample, such as sample mean $\bar{x}$, sample variance s^2 , and sample standard deviation s .

Main types of inferential statistics: point estimation, interval estimation, and hypothesis testing.

2. Meaning of Hypothesis Testing

A hypothesis is an assumption or claim about a population parameter. Hypothesis testing is a statistical method used to decide whether there is enough evidence to reject a claim about a population.

Example: Claim - average customer satisfaction score is above 4. We test this using sample data.

3. Null and Alternative Hypothesis

Term	Meaning	Example
Null Hypothesis (H_0)	Default assumption or claim.	$H_0: \mu = 50$
Alternative Hypothesis (H_1 / H_a)	Claim supported if H_0 is rejected.	$H_1: \mu \neq 50$

Important rule: equality signs usually belong to the null hypothesis.

$=, \leq, \geq \rightarrow H_0$ | $\neq, <, > \rightarrow H_1$

4. Significance Level and Confidence Level

Significance level (α) is the probability of rejecting H_0 when H_0 is actually true. Common values are 0.05, 0.01, and 0.10.

Confidence Level = $1 - \alpha$

Significance Level α	Confidence Level
0.01	99%
0.05	95%
0.10	90%

5. Type I and Type II Errors

Error Type	Meaning	Also Called	Probability
Type I Error	Rejecting H_0 when H_0 is actually true.	False Positive	α
Type II Error	Failing to reject H_0 when H_0 is actually false.	False Negative	β

Power of test = $1 - \beta$. It means the probability of correctly rejecting a false null hypothesis.

6. Parametric and Non-Parametric Tests

Type	Meaning	Examples
Parametric Tests	Assume a specific distribution, usually normal distribution. Use mean, standard deviation, and variance.	Z-test, t-test, F-test, ANOVA
Non-Parametric Tests	Do not assume normal distribution. Often use ranks, signs, or frequencies.	Chi-square test, Mann-Whitney U-test, Wilcoxon test, Kruskal-Wallis test

7. One-Tailed and Two-Tailed Tests

Test Type	When Used	Hypotheses
Right-Tailed Test	To check whether value is greater.	$H_0: \mu \leq \mu_0$, $H_1: \mu > \mu_0$
Left-Tailed Test	To check whether value is smaller.	$H_0: \mu \geq \mu_0$, $H_1: \mu < \mu_0$
Two-Tailed Test	To check whether there is any difference in either direction.	$H_0: \mu = \mu_0$, $H_1: \mu \neq \mu_0$

8. p-Value

The p-value is the probability of getting the observed result, or something more extreme, assuming H_0 is true. It tells how strong the evidence is against H_0 .

**If p-value $\leq \alpha \rightarrow$ Reject H_0
If p-value $> \alpha \rightarrow$ Fail to reject H_0**

Example: p-value = 0.012 and $\alpha = 0.05$. Since $0.012 < 0.05$, reject H_0 .

9. Critical Value Approach

A critical value is a boundary value. If the test statistic falls in the rejection region, reject H_0 .

α	One-Tailed Z	Two-Tailed Z
0.10	1.28	± 1.645
0.05	1.645	± 1.96
0.01	2.33	± 2.576

- Right-tailed test: Reject H_0 if $Z \geq Z_\alpha$.
- Left-tailed test: Reject H_0 if $Z \leq -Z_\alpha$.

- Two-tailed test: Reject H_0 if $Z < -Z_{\alpha/2}$ or $Z > Z_{\alpha/2}$.

10. Confidence Interval Approach

If the hypothesized value is outside the confidence interval, reject H_0 . If it is inside the confidence interval, fail to reject H_0 .

Example: 95% CI = (3.5, 4.5), $H_0: \mu = 5$. Since 5 is outside the interval, reject H_0 .

11. Steps of Hypothesis Testing

- 1 State H_0 and H_1 .
- 2 Choose significance level α .
- 3 Select the appropriate test.
- 4 Calculate the test statistic.
- 5 Find p-value or critical value.
- 6 Make the decision: reject H_0 or fail to reject H_0 .
- 7 Write conclusion in the problem context.

12. Which Test to Use?

Situation	Test
Large sample and population standard deviation known	Z-test
Small sample or population standard deviation unknown	t-test
Compare sample mean with population mean, σ unknown	One-sample t-test
Compare two independent group means	Independent two-sample t-test
Compare before and after values	Paired t-test
Categorical or frequency data	Chi-square test
Compare variances	F-test
Compare means of 3 or more groups	ANOVA
Two independent groups with non-normal data	Mann-Whitney U-test

13. t-Test

A t-test is used to compare means when sample size is small, population standard deviation is unknown, and data is approximately normal.

Types of t-test: one-sample t-test, independent two-sample t-test, and paired t-test.

One-Sample t-Test

Used to compare a sample mean with a known or hypothesized population mean.

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

where $\bar{x}$ = sample mean, μ_0 = hypothesized mean, s = sample standard deviation, n = sample size. Degrees of freedom: $df = n - 1$.

Example: One-Sample t-Test

A company claims bulbs last 1000 hours. Given $n = 25$, $\bar{x} = 980$, $s = 60$, $\alpha = 0.05$.

$$H_0: \mu = 1000, H_1: \mu \neq 1000$$

$$t = (980 - 1000) / (60 / \sqrt{25}) = -20 / 12 = -1.667$$

$df = 24$, critical $t = \pm 2.064$. Since $|-1.667| < 2.064$, fail to reject H_0 . Conclusion: insufficient evidence that bulb life differs from 1000 hours.

Independent Two-Sample t-Test

Used to compare means of two independent groups, such as marks of students taught by Method A and Method B.

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$df = n_1 + n_2 - 2$$

Paired t-Test

Used when the same group is measured two times, such as before and after treatment, training, or diet.

$$H_0: \mu_d = 0$$

$$H_1: \mu_d \neq 0$$

$$t = \bar{d} / (sd / \sqrt{n})$$

$$df = n - 1$$

14. Z-Test

A z-test is used when sample size is large, population standard deviation σ is known, and data is approximately normal.

$$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

Z-test can be used for one-sample mean test, two-sample mean test, and proportion test.

15. Chi-Square Test

The chi-square test is a non-parametric test used for categorical data. It checks whether observed frequencies are significantly different from expected frequencies.

$$\chi^2 = \sum [(O - E)^2 / E]$$

Conditions: data should be frequency/count data, observations should be independent, expected frequency should usually be at least 5, and categories should be mutually exclusive.

Chi-Square Goodness of Fit Test

Used to check whether observed distribution matches expected distribution. Example: testing whether a die is fair.

$$H_0: \text{Observed frequencies match expected distribution}$$

$$H_1: \text{Observed frequencies do not match expected distribution}$$

$$df = k - 1$$

Example: A die is rolled 120 times. Expected frequency for each face = $120/6 = 20$. Observed frequencies: 18, 22, 17, 24, 19, 20.

$$\chi^2 = 0.2 + 0.2 + 0.45 + 0.8 + 0.05 + 0 = 1.70$$

df = 5, critical value at $\alpha = 0.05$ is 11.07. Since $1.70 < 11.07$, fail to reject H_0 . The die appears to be fair.

Chi-Square Test of Independence

Used to check whether two categorical variables are related, such as gender and product preference, or treatment type and recovery status.

H_0 : The two variables are independent

H_1 : The two variables are associated

Expected Frequency: $E_{ij} = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$

$df = (r - 1)(c - 1)$

16. Mann-Whitney U-Test

The Mann-Whitney U-test, also called the Wilcoxon Rank-Sum Test, is a non-parametric test used to compare two independent groups.

- Use when data is not normally distributed.
- Use when sample size is small.
- Use when data is ordinal or ranked.
- It compares ranks instead of means.

H_0 : Both groups have the same distribution

H_1 : One group tends to have larger or smaller values

17. ANOVA

ANOVA stands for Analysis of Variance. It is used to compare means of three or more groups.

Example: comparing marks of students taught by three teaching methods.

Why not multiple t-tests? If we compare 3 groups separately at $\alpha = 0.05$, the probability of at least one Type I error becomes $1 - (0.95)^3 = 14.3\%$. ANOVA controls this error.

One-Way ANOVA

Used when there is one independent variable, one continuous dependent variable, and three or more groups.

$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$

H_1 : At least one group mean is different

Important: ANOVA tells that a difference exists, but it does not tell exactly which groups are different. For that, post-hoc tests are used.

ANOVA Logic and Formula

ANOVA divides total variation into between-group variation and within-group variation.

$SST = SSB + SSW$

$F = MSB / MSW$

Source	Sum of Squares	df	Mean Square	F
Between Groups	SSB	$k - 1$	$MSB = SSB / (k - 1)$	MSB / MSW
Within Groups	SSW	$N - k$	$MSW = SSW / (N - k)$	

Total	SST	N - 1		
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ANOVA Example

Three teaching methods are tested:

Method A: 70, 75, 80, 65, 70

Method B: 85, 90, 88, 92, 80

Method C: 60, 65, 70, 55, 65

Means: A = 72, B = 87, C = 63, Grand mean = 74.

$$SSB = 5[(72 - 74)^2 + (87 - 74)^2 + (63 - 74)^2] = 1470$$

$$\text{Given SSW } 380. F = (1470 / 2) / (380 / 12) = 735 / 31.67 = 23.2$$

Critical $F(2,12) = 3.89$. Since $23.2 > 3.89$, reject H_0 . Conclusion: teaching methods differ significantly.

Types of ANOVA

Type	Meaning	Checks
One-Way ANOVA	One factor.	Effect of one independent variable on a continuous dependent variable.
Two-Way ANOVA	Two factors.	Main effect of Factor A, main effect of Factor B, and interaction effect $A \times B$.

Effect Size in ANOVA

Effect size tells how strong the difference is. A common measure is eta squared.

$$\eta^2 = SSB / SST$$

η^2 Value	Effect
0.01	Small
0.06	Medium
0.14	Large

18. Scenario-Based Examples

Scenario 1: Coffee Shop Waiting Time

Claim: average customer waits less than 5 minutes. A customer records wait times: sample mean = 5.4, standard deviation = 1, sample size = 30, $\alpha = 0.05$.

$$H_0: \mu = 5$$

$$H_1: \mu > 5$$

This is a right-tailed t-test because population standard deviation is unknown. Given p-value = 0.018. Since $0.018 < 0.05$, reject H_0 . There is enough evidence that average wait time is more than 5 minutes.

Scenario 2: Customer Satisfaction Proportion

Claim: 80% customers are satisfied. Survey: $n = 100$, satisfied customers = 72, so $\hat{p} = 72/100 = 0.72$.

$$H_0: p = 0.80$$

$$H_1: p < 0.80$$

This is a left-tailed z-test for proportion. Given p-value = 0.0228. Since $0.0228 < 0.05$, reject H_0 . There is evidence that customer satisfaction is less than 80%.

19. Important Correction to Remember

If a research hypothesis says a new fuel injection system should increase fuel efficiency above 40 km/L, the correct hypotheses are:

$$H_0: \mu \leq 40$$

$$H_1: \mu > 40$$

Because the research claim is that average fuel efficiency exceeds 40 km/L, this is a right-tailed test.

20. Short Revision Notes

- Hypothesis testing = decision-making using sample data.
- H_0 = default claim.
- H_1 = claim we want to support.
- α = significance level.
- p-value = probability of observed result assuming H_0 is true.
- If p-value $\leq \alpha$, reject H_0 .
- Type I error = rejecting true H_0 .
- Type II error = failing to reject false H_0 .
- Z-test = large sample or known σ .
- t-test = small sample or unknown σ .
- Chi-square = categorical/frequency data.
- ANOVA = compare 3 or more means.
- U-test = non-parametric test for two independent groups.

21. Most Important Formulas

Concept	Formula
Z-Test	$Z = (\bar{x} - \mu) / (\sigma / n)$
One-Sample t-Test	$t = (\bar{x} - \mu_0) / (s / n)$
Paired t-Test	$t = d / (sd / n)$
Chi-Square Test	$\chi^2 = \sum [(O - E)^2 / E]$
Expected Frequency	$E_{ij} = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$
ANOVA F-Test	$F = MSB / MSW$
Eta Squared	$\eta^2 = SSB / SST$

22. Final Exam Writing Format

- 1 State H_0 and H_1 .
- 2 Identify the test type.
- 3 Write significance level α .
- 4 Write the formula.
- 5 Calculate the test statistic.

- 6 Compare with critical value or p-value.
- 7 Decision: reject or fail to reject H_0 .
- 8 Final conclusion in words.

This format works for Z-test, t-test, chi-square test, ANOVA, and proportion test.